RAJALAKSHMI ENGINEERING COLLEGE

Teja Shree U 251001282-2025-IT

T2

Started on	Friday, 31 July 2026, 12:15 PM
State	Finished
Completed on	Friday, 31 July 2026, 12:27 PM
Time taken	11 mins 46 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using the counter method.

```
void function (int n)
{
    int i= 1;
```

```
    int s =1;
```

```
    while(s <= n)
    {
         i++;
         s += i;
     }
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

For example:

Input	Result
9	12

Answer: (penalty regime: 0 %)

```c
#include<stdio.h>
void function(int n){
    int i=1;
    int s=1;
    int counter=0;
    counter+=2;
    while(s<=n){
        counter++;
        i++;
        counter++;
        s+=i;
        counter++;
    }
    counter++;
    printf("%d",counter);
}
int main(){
    int n;
    scanf("%d",&n);
    function(n);
    return 0;
}
```

	Input	Expected	Got	
✔	9	12	12	✔
✔	4	9	9	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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Teja Shree U 251001282-2025-IT

T2

Started on	Friday, 31 July 2026, 12:27 PM
State	Finished
Completed on	Friday, 31 July 2026, 12:57 PM
Time taken	30 mins 34 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using the counter method.

```
void func(int n)
{
    if(n==1)
    {
      printf("*");
    }
    else
    {
     for(int i=1; i<=n; i++)
     {
       for(int j=1; j<=n; j++)
       {
          printf("*");
          printf("*");
          break;
       }
     }
   }
 }
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer: (penalty regime: 0 %)

```
#include<stdio.h>
void func(int n){
    int counter=0;
    if(n==1){
    counter++;
   // printf(" ");
    counter++;
    }
    else{
        counter++;
        for(int i=1;i<=n;i++){
            counter++;
                for(int j=1;j<=n;j++){
                    counter++;
                   // printf(" ");
                    counter++;
                    //printf(" ");
                    counter++;
                    break;
                }
                counter++;
        }
        counter++;
    }
   printf("%d",counter) ;
}

int main(){
```

```
29      int n;
30      scanf("%d",&n);
31      func(n);
32      return 0;
33  }
34
```

	Input	Expected	Got	
✔	2	12	12	✔
✔	1000	5002	5002	✔
✔	143	717	717	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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RAJALAKSHMI ENGINEERING COLLEGE

Teja Shree U 251001282-2025-IT

T2

Started on	Friday, 31 July 2026, 12:58 PM
State	Finished
Completed on	Friday, 31 July 2026, 1:19 PM
Time taken	21 mins 34 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using counter method.

```
Factor(num) {
{
   for (i = 1; i <= num;++i)
   {
    if (num % i== 0)
       {
         printf("%d ", i);
       }
    }
 }
```

Note: No need of counter increment for declarations and scanf() and counter variable printf() statement.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

```c
#include<stdio.h>
void factor(int n){
    int counter=0;

    for(int i=1;i<=n;i++){
        counter++;
        if(n%i==0){
            //printf("%d",i);
            counter++;
        }
        counter++;
    }
    counter++;
    printf("%d",counter);

}
int main(){
    int n;
    scanf("%d",&n);
    factor(n);

}
```

	Input	Expected	Got	
✔	12	31	31	✔
✔	25	54	54	✔
✔	4	12	12	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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RAJALAKSHMI ENGINEERING COLLEGE

Teja Shree U 251001282-2025-IT

T2

Started on	Friday, 7 August 2026, 8:25 AM
State	Finished
Completed on	Tuesday, 25 August 2026, 12:08 AM
Time taken	17 days 15 hours
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using counter method.

```
void function(int n)
{
    int c= 0;
    for(int i=n/2; i<n; i++)
        for(int j=1; j<n; j = 2 * j)
            for(int k=1; k<n; k = k * 2)
                c++;
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

```c

#include <stdio.h>

void function(int n)
{
    int count = 0;

    count++;  // outer loop initialization

    for (int i = n / 2; i < n; i++)
    {
        count++;  // i++

        count++;  // middle loop initialization

        for (int j = 1; j < n; j = 2 * j)
        {
            count++;  // j++

            count++;  // inner loop initialization

            for (int k = 1; k < n; k = k * 2)
            {
                count++;  // k++
                count++;  // c++
            }
        }
    }

    count++;  // return

    printf("%d\n", count);
}

int main()
{
    int n;
    scanf("%d", &n);
```

```
39
40      function(n);
41
42      return 0;
43  }
44
45
```

	Input	Expected	Got	
✔	4	30	30	✔
✔	10	212	212	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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RAJALAKSHMI ENGINEERING COLLEGE

Teja Shree U 251001282-2025-IT

T2

Started on	Monday, 24 August 2026, 11:57 PM
State	Finished
Completed on	Tuesday, 25 August 2026, 12:03 AM
Time taken	5 mins 41 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using counter method.

```
void reverse(int n)
{
   int rev = 0, remainder;
   while (n != 0)
    {
        remainder = n % 10;
        rev = rev * 10 + remainder;
        n/= 10;

    }
print(rev);
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

```c
#include <stdio.h>

void reverse(int n)
{
    int rev = 0, remainder;
    int count = 0;

    while (n != 0)
    {
        count++;                    // while condition/iteration

        remainder = n % 10;
        count++;

        rev = rev * 10 + remainder;
        count++;

        n /= 10;
        count++;
    }

    count += 3;                     // final while-condition checks

    printf("%d\n", count);
}

int main()
{
    int n;
    scanf("%d", &n);

    reverse(n);

    return 0;
}
```

	Input	Expected	Got	
✔	12	11	11	✔
✔	1234	19	19	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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